

JAVA Assignment

Spring Boot Framework should be used to implement the interview assignment.

Requirements

- Candidates are expected to implement the required features based on the provided Scenario and User Stories.
- You should create a new Java Application and deliver the code, APIs and API documentation. (Feel free to use the database of your choice, and you may use any frameworks / submodules within Spring Boot)
- Each implementation should be equipped with unit tests.
- A Read-Me file should be provided which describes the details to run the program and setup database.
- Once complete, please place your project in a zipped folder (.zip) and send it back to the recruiter for the team to review.

Bonus points: If you can use Docker/Docker-Compose to run your application.

Scenario

Create a pricing algorithm for BTC

The price starts from 100 USD and increases by 10 USD every 5 seconds for the first 3 minutes. At the 3 min period, it should reach 460 USD. Then it should decrease by 10 USD every 5 seconds for the next 3 minutes. At the 6 min period, the price is 100 USD. This cycle keeps repeating continuously.

Application User Stories:

1. As a user, I want to be able to add new users for the platform by name and email address.
2. As a user, I want to receive 1,000 USD in my account when I create a new user account successfully.
3. As a user, I want to be able to buy and sell BTC at the current price.
4. As a user, I want to be able to see my transaction history, which includes the price of BTC and the balance of BTC and USD.
5. As a user, I want to be able to delete the existing users.